Experiment:18 Simulation and Performance Evaluation of a Millimeter-Wave Massive-MIMO Communication Link with Beamforming and Spatial Multiplexing

Aim: To simulate a millimeter-wave Massive-MIMO communication link using MATLAB and evaluate its spectral efficiency, bit error rate (BER), and beamforming performance with spatial multiplexing.

Objective:

To simulate a millimeter-wave Massive-MIMO communication link and evaluate spectral efficiency, bit error rate (BER), and beamforming performance using MATLAB.

Apparatus:

Computer/Laptop

Matlab Software

Procedure:

- Open MATLAB and create a new script.
- Define the carrier frequency, antenna numbers, spatial streams, and SNR range.
- Generate a MIMO channel matrix for the communication link.
- Apply singular value decomposition (SVD) to obtain spatial transmission and reception directions.
- Simulate transmitted data symbols and add noise for different SNR values.
- Calculate spectral efficiency and bit error rate.
- Plot spectral efficiency and BER against SNR.
- Observe the results and evaluate the communication system's performance.

Code:

```
clc;

clear;

close all;

% System parameters

Nt = 16;

Nr = 4;

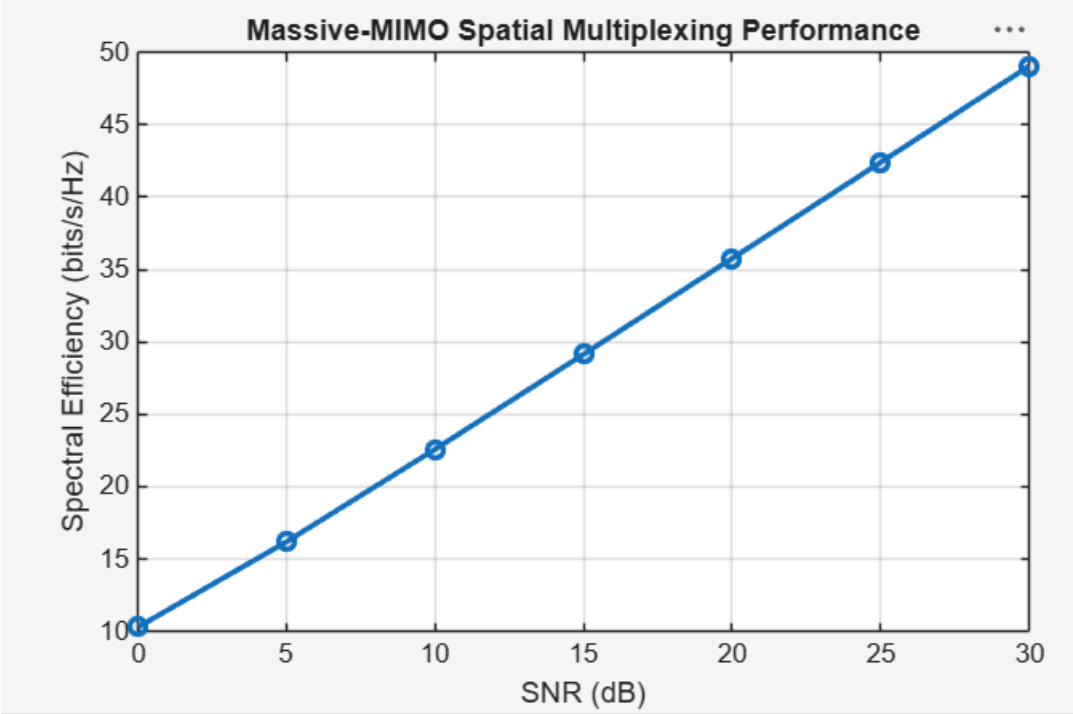
Ns = 4;

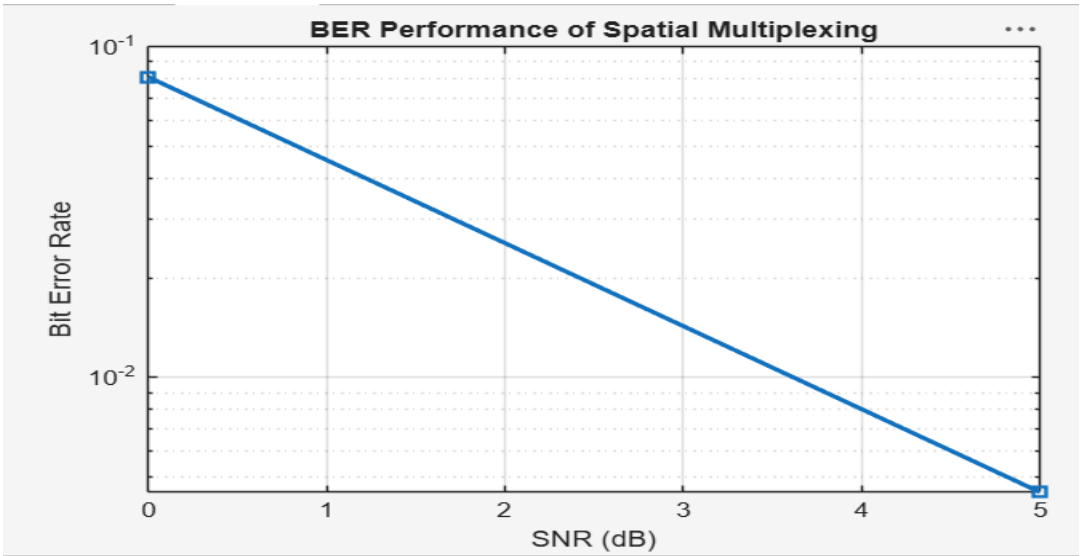
fc = 28e9;
```

```
c = 3e8;
lambda = c/fc;
SNR_dB = 0:5:30;
Nsym = 1000;
% Antenna arrays
tx = (0:Nt-1)';
rx = (0:Nr-1)';
% Channel matrix
H = (randn(Nr,Nt) + 1j*randn(Nr,Nt))/sqrt(2);
% SVD-based spatial multiplexing
[U,S,V] = svd(H);
H_eff = U(:,1:Ns)'*H*V(:,1:Ns);
capacity = zeros(size(SNR_dB));
ber = zeros(size(SNR_dB));
for k = 1:length(SNR_dB)
    snr = 10^(SNR_dB(k)/10);
    capacity(k) = real(log2(det(eye(Ns) + ...
        (snr/Ns)*(H_eff'*H_eff))));
    bits = randi([0 1],Ns,Nsym);
    symbols = 2*bits-1;
    symbols = symbols/sqrt(1);
    noise = sqrt(1/(2*snr)) * ...
        (randn(Ns,Nsym)+1j*randn(Ns,Nsym));
    received = symbols + noise;
    detected = real(received) > 0;
    ber(k) = sum(detected(:) ~= bits(:))/numel(bits);
end
```

```
% Spectral efficiency
figure;
plot(SNR_dB,capacity,'-o','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Spectral Efficiency (bits/s/Hz)');
title('Massive-MIMO Spatial Multiplexing Performance');
% BER performance
figure;
semilogy(SNR_dB,ber,'-s','LineWidth',2);
grid on;
xlabel('SNR (dB)');
ylabel('Bit Error Rate');
title('BER Performance of Spatial Multiplexing');
% Beamforming gain
beamforming_gain = Nt*Nr;
fprintf('Carrier Frequency = %.1f GHz\n',fc/1e9);
fprintf('Wavelength = %.4f m\n',lambda);
fprintf('Transmit Antennas = %d\n',Nt);
fprintf('Receive Antennas = %d\n',Nr);
fprintf('Spatial Streams = %d\n',Ns);
fprintf('Ideal Array Gain = %.2f dB\n', ...
    10*log10(beamforming_gain));
```

Output:





Result:

The Massive-MIMO communication link is simulated in MATLAB, and its spectral efficiency and BER are evaluated under different SNR conditions.